

High dimensional geometry and Regularization

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This week

High dimensional Gaussians

Geometry in high dimensions

Gaussian Mixture Modeling

Dimensionality reduction (PCA/JL)

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High dimensional Gaussians

- Geometry in high dimensions

- Gaussian Mixture Modeling

- Dimensionality reduction (PCA/JL)

Ridge and Lasso

- Explanations

- Compressive sensing

- Matrix norms

High dimensional Gaussian

Multivariate Gaussian

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{d}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left(-(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right)$$

$$\mu = \mathbb{E}X \text{ (mean)}$$

$$\Sigma = \mathbb{E}(X - \mu)(X - \mu)^T \text{ (covariance)}$$

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$\Sigma = \mathbb{E}(X - \mu)(X - \mu)^T$ (covariance)

Where is the probability concentrated?

Concentration of measure

$$U \sim N(\mu, \sigma^2 I)$$

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Thin shell with width $\sqrt{d}$

For all $\delta > 0$,

$$\mathbf{P} \left(\|U - \mu\|^2 \leq \sigma^2 \left(d + 2\sqrt{d \ln \frac{1}{\delta}} \right) \right) \geq 1 - \delta$$

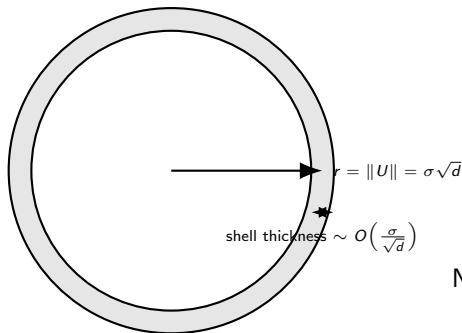
and

$$\mathbf{P} \left(\|U - \mu\|^2 \geq \sigma^2 \left(d - 2\sqrt{d \ln \frac{1}{\delta}} \right) \right) \geq 1 - \delta$$

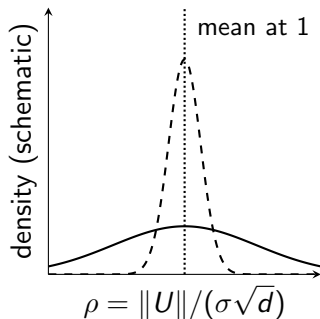
Gaussian norm concentrates in a thin shell

Let $U \sim \mathcal{N}(0, \sigma^2 I_d)$.

Most probability mass
lives in the thin annulus



2D cross-section of a high- d
phenomenon



Normalized radius tightens as d
grows

Concentration of measure

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Around equator relative to any unit vector $\mathbf{z}$

For all $\delta > 0$,

$$\mathbf{P} \left(\mathbf{z}^T (U - \mu) \leq \sigma \sqrt{2 \ln \frac{1}{\delta}} \right) \geq 1 - \delta$$

Learning mixtures of Gaussians

Cluster n points in $\mathbb{R}^d$ into k clusters

Powerful and flexible model: Gaussian mixtures

$$X \sim \sum_{i=1}^k \pi_i \mathcal{N}(\mu_i, \Sigma_k)$$

Note: even common covariance $\Sigma_k = \Sigma$ versatile

Clustering in low dimensions, few clusters

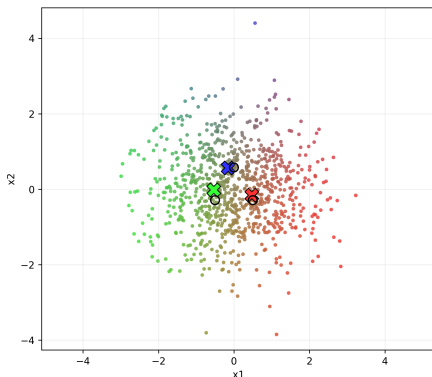
k —means

- choose centers $\mu_1, \dots, \mu_k$ at random
- assign each example to nearest mean
- update centers and repeat prior step till convergence

Soft version: Expectation Maximization

- Fits most likely GMM iteratively
- For Gaussians, soft version of k —means

EM for 2D Gaussian Mixtures (known $\sigma^2 I$, $K = 3$)



Post EM training

X = estimated, o = true

Color of point: mix of red/blue/green representing affinity to corresponding cluster

Setup (latent-variable model)

Observations $X_n \in \mathbb{R}^2$

Latent labels

$$Z_n \in \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$\mathbf{P}(X_n, Z_n) = \mathbf{P}(Z_n)f(X_n | Z_n)$$

with $f(\cdot | Z_n) \sim \mathcal{N}(\mu_k, \sigma^2 I)$

$$\mathbf{P}(Z_n = [k]) \stackrel{\text{def}}{=} \pi_k$$

Don't know π or μ

To find out: maximize likelihood

$$\max_{\mu, \pi} \log \mathbf{P}(X_1, \dots, X_N | \mu, \pi)$$

Model and notation

Parameters

Mixing weights $\pi_k \geq 0$, $\sum_k \pi_k = 1$;
means $\mu_k \in \mathbb{R}^2$;
known covariance $\sigma^2 I$.

Use Θ for vector of all parameters above
Joint and marginal:

$$\mathbf{P}(X_n, Z_n | \Theta) = \mathbf{P}(X_n | Z_n, \Theta) \mathbf{P}(Z_n | \Theta)$$

$$\mathbf{P}(X_n | \Theta) = \sum_{k=1}^3 \mathbf{P}(X_n, Z_n = k | \Theta).$$

Complete-data log-likelihood

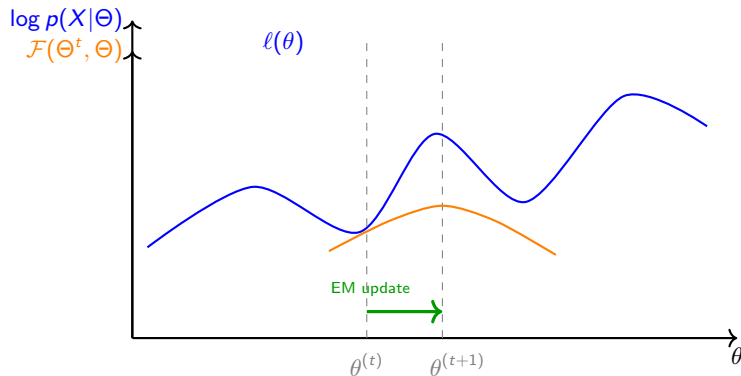
If we “knew” Z_n , fitting parameters in each cluster is finding the mean of a Gaussian from its samples. But because we do not know cluster identity, we maximize

$$\log \mathbf{P}(X_1, \dots, X_N | \Theta) = \log \prod p(X_i | \Theta)$$

Expectation-Maximization

$\log \mathbf{P}(X_1, \dots, X_N | \Theta)$ is not a simple convex function of Θ . E-M is an iterative category of algorithms to find the value of Θ that maximize log-likelihood.

Evidence Lower Bound (ELBO)



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If Θ^t is our parameter guess at time t , the evidence lower bound (ELBO) satisfies

$$\begin{aligned}\mathcal{F}(\Theta^t, \Theta) &\leq \log p(X|\Theta) \quad \text{for all } \Theta \\ \mathcal{F}(\Theta^t, \Theta^t) &= \log p(X|\Theta^t)\end{aligned}$$

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If Θ^{t+1} maximizes $\mathcal{F}$ over all Θ , then

$$\log p(X|\Theta^{t+1}) \geq \log p(X|\Theta^t)$$

Formulating $\mathcal{F}$ is the E-step:

$$\mathcal{F}(\Theta^t, \Theta) = \mathbb{E}[\log p(X, Z|\Theta) \mid X, \Theta^t] - \mathbb{E}[\log p(Z|X, \Theta^t) \mid X, \Theta^t]$$

Gaussian Mixture Modeling

What is $q(Z) \stackrel{\text{def}}{=} \mathbf{P}(Z|X, \Theta^t) = \prod_n \mathbf{P}(Z_n|X_n, \Theta^t)$?

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$$q_n([k]|X_n) = \mathbf{P}(Z_n = [k]|X_n, \Theta^t) = \frac{\pi_k^{(t)} f(X_n | \mu_k^{(t)}, \sigma^2 I)}{\sum_j \pi_j^{(t)} f(X_n | \mu_j^{(t)}, \sigma^2 I)}$$

where $f(\cdot|\mu, \Sigma)$ is the Gaussian density with μ and Σ .

Now,

$$\mathbb{E}[\log p(X, Z|\Theta) | X, \Theta^t] = \mathbb{E}[\log p(Z|\Theta) | X, \Theta^t] + \mathbb{E}[\log p(X|Z, \Theta) | X, \Theta^t].$$

Because our parameters match this split

As before calling $\mathbf{P}(Z_n|X_n, \Theta^t) = q_n$, and $q = \prod_n q_n$ we need to maximize over π, μ

$$\mathbb{E}_q[\log p(Z|\Theta)] + \mathbb{E}_q[\log p(X|Z, \Theta)],$$

First term only depends on π , second only on μ .

Gaussian Mixture modeling: term 1

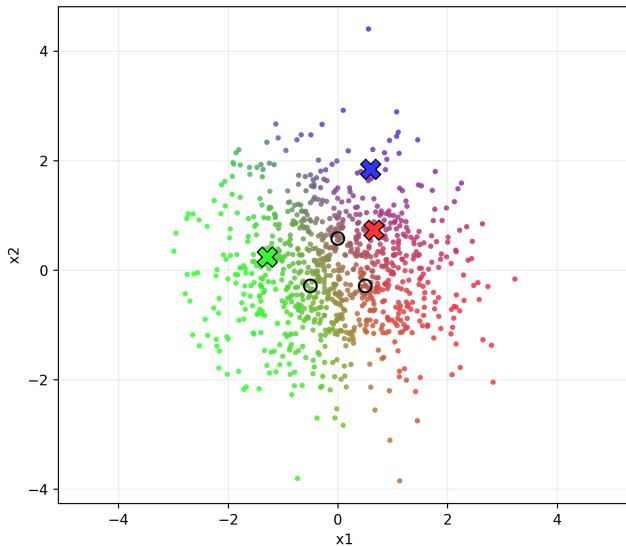
We need to maximize over all choices of $\pi = \begin{bmatrix} \pi_1 \\ \pi_2 \\ \pi_3 \end{bmatrix}$,

$$\mathbb{E}_q \sum_n Z_n^T \log \pi = \sum_n (\mathbb{E}_{q_n} Z_n)^T \log \pi = \left(\sum_n \mathbb{E}_{q_n} Z_n \right)^T \log \pi$$

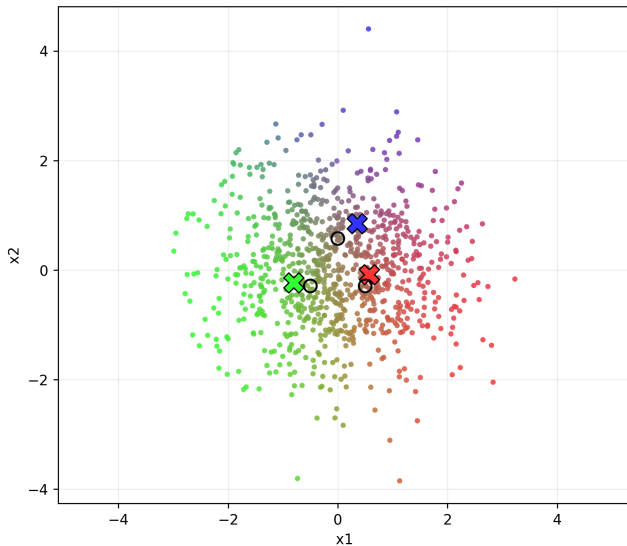
How to maximize above? Make π proportional to $(\sum_n \mathbb{E}_{q_n} Z_n)$

Similarly for term 2 (HW)

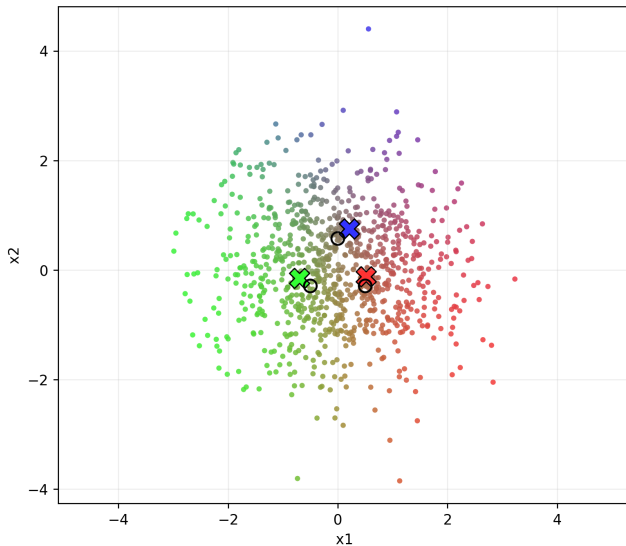
EM snapshot: step 0



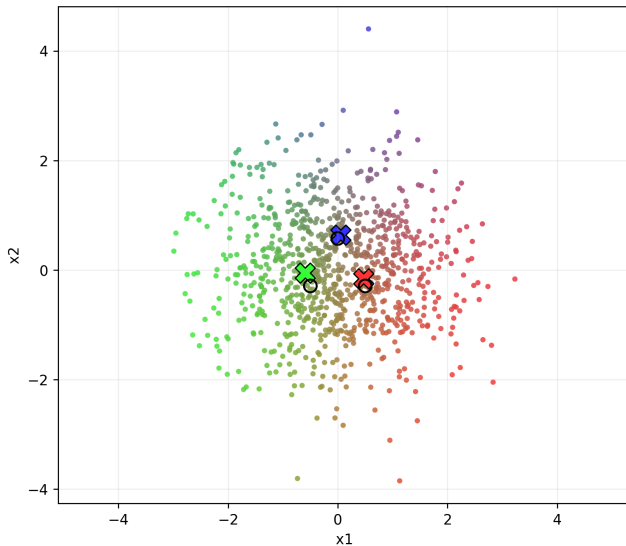
EM snapshot: step 1



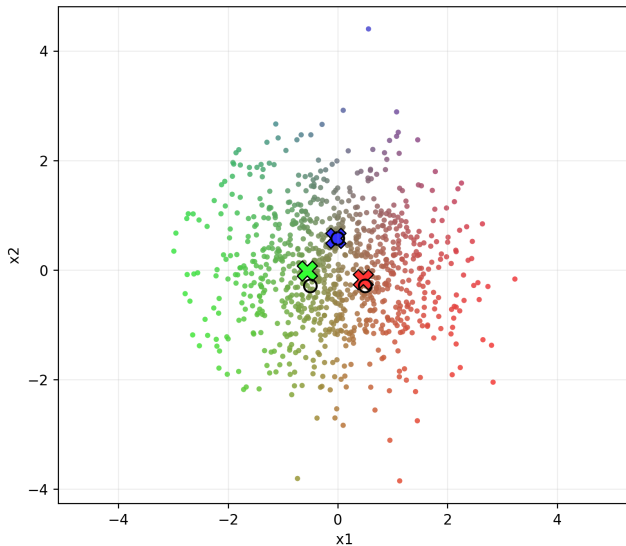
EM snapshot: step 2



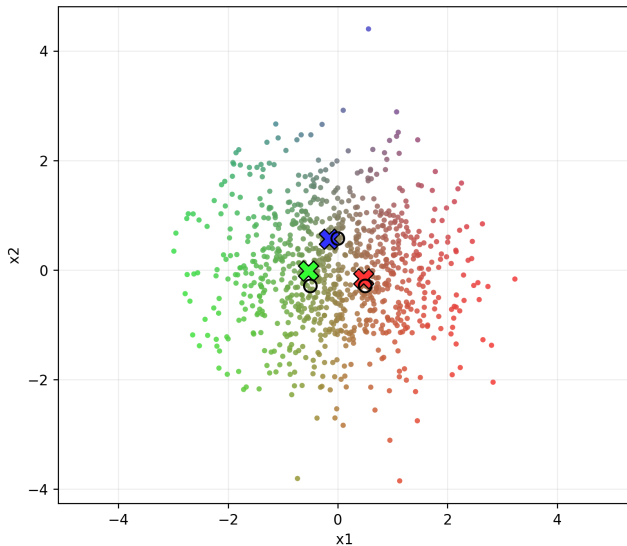
EM snapshot: step 5



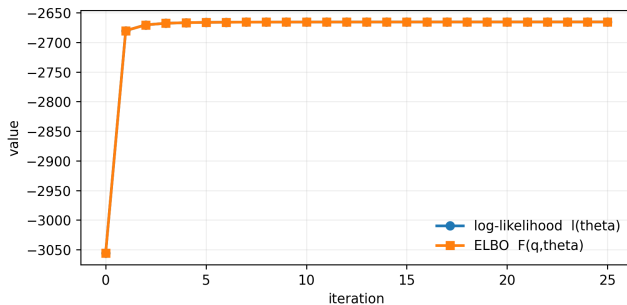
EM snapshot: step 10



EM snapshot: step 25



Objective tracking: ELBO and log-likelihood



Where ELBO shows up again

ELBO is the unifying pattern

- **Variational inference / VAEs:** maximize an ELBO when the exact posterior is intractable.
- **Diffusion models:** treat the noising trajectory as a latent variable; training often comes from a variational lower bound on $\log p_\theta(x_0)$ that decomposes into per-timestep KL terms.
- **Interpretation:** “maximize likelihood” $\Rightarrow$ “maximize a tractable lower bound” $\Rightarrow$ algorithmic updates (EM / VI / diffusion losses).

Diffusion sketch (see later)

Forward (fixed) process $q(x_{1:T} | x_0)$; reverse model $p_\theta(x_{t-1} | x_t)$. A standard bound has the form

$$\log p_\theta(x_0) \geq \mathbb{E}_q \left[\log p(x_T) + \sum_{t=1}^T \log p_\theta(x_{t-1} | x_t) - \sum_{t=1}^T \log q(x_t | x_{t-1}) \right],$$

which becomes a sum of KL / denoising terms across timesteps under common parameterizations.

High Dimensional GMM

Several modern systems use GMM explicitly and implicitly on high-d Gaussians

Explicit:

- Speech Recognition (GMM model MPCC)

- Visual features (gradients with respect to GMM models of SIFT descriptors)

- Motion Detection, Robotics Vision

- Biomedical (gene expression clustering)

Implicit:

- Diffusion models (Stable diffusion, Dall-E)

- Variational Autoencoders (anomaly detection)

Why everywhere?

Any smooth density can be approximated arbitrarily well by a Gaussian mixture

In high dimensions

Recall: most probability in $\mathcal{N}(\mu, \sigma^2 I)$ close to $\sigma\sqrt{d}$.

$$\mathbf{P} \left(\|X - \mu\|^2 \geq \sigma^2 \left(d - 2\sqrt{d \ln \frac{1}{\delta}} \right) \right) \geq 1 - \delta$$

Probability of finding a point near μ is $\exp(-\mathcal{O}(d))$

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Need $\exp(\mathcal{O}(d))$ points to even have a point $\leq \frac{1}{2}\sigma\sqrt{d}$!

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Most plausible data sizes: “few scattered specks of dust in an enormous void” (Dasgupta '99)

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Low dim algorithms need exponential in d examples

Yet...

High-d Gaussian clustering is actually easier!

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Blessing of dimensionality

(See board)

Key idea: Project into few dimensions

PCA (not such a great idea)

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Computationally expensive, but recovers high variance directions for full mixture

High variance direction may not necessarily be informative for cluster separation

it is possible PCA collapses components of the mixture on top of each other (or nearly so)

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Alternatively random projections (Johnson-Lindenstrauss):

$\frac{1}{\epsilon^2} \log n$ projections retain all pairwise distances

projected space still too large

exponential in $\frac{1}{\epsilon^2} \log n$ is $n^{\frac{1}{\epsilon^2}}$

Johnson Lindenstrauss Lemma

Random projections preserve pairwise distances

For any ϵ and integer n , let $k = \frac{8 \ln n}{\epsilon^2}$. For all $\mathbf{z}_1, \dots, \mathbf{z}_n \in \mathbb{R}^d$, there exists $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ such that for all pairs $\mathbf{z}_i, \mathbf{z}_j$

$$\|f(\mathbf{z}_i) - f(\mathbf{z}_j)\|^2 \in (1 \pm \epsilon) \|\mathbf{z}_i - \mathbf{z}_j\|^2$$

These f can simply be random projections!

Applications of JL lemma

Approximate Nearest Neighbor (vector databases for LLM embeddings, Faiss, ScaNN)

Sketching and streaming algorithms (network traffic, Apache Flink)

In fact: Project into even fewer dimensions

- Don't worry about retaining all pairwise distances
 - $O(\log k)$ projections (k -clusters) (remember assignment 1)
 - retain distances between means
 - push points closer to mean in each cluster!

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 $\mathcal{O}(\log k)$ projections (k -clusters) (remember assignment 1)
retain distances between means
push points closer to mean in each cluster!

Series of recent results on several common examples in the low-d space, how they recover parameters in high-d space

Distance between means

If $\|\mu_1 - \mu_2\| > \Omega(d^{1/4})$, should expect to separate out clusters

Note that in this regime, the spheres are not disjoint

Yet we should expect all points in one cluster to be closer to each other than points in other clusters

Why Gaussian mixtures

In principle, GMs can model any continuous distribution

Commonly used as explorative analysis or to infer clusterings

Few particular examples (potential projects):

- Clustering ECGs

- Training LLMs

Gaussian random matrices

If A is a $k \times n$ matrix, entries iid Gaussian
rows, cols independently chosen Gaussian multivariate
satisfy something called the Restricted isometry property
all small subset of columns approximately orthogonal

Key property used in Compressed Sensing
extends the Shannon-Nyquist theorem
used to shorten MRI acquisition on conventional equipment,
network tomography, radio astronomy and optical interferometry
(aperture synthesis)

Compressed Sensing

If $\mathbf{x}^*$ is a S -sparse signal in $\mathbb{R}^n$

$\mathbf{y} = A\mathbf{x}^*$ (ie k linear measurements of $\mathbf{x}$)

If k is very small, can we still find $\mathbf{x}^*$?

Compare with Shannon-Nyquist sampling

Convex relaxation

$y = A\mathbf{x}$ is underdetermined

Convex relaxation

$y = Ax$ is underdetermined
infinite solutions
which solution to choose?

Finding sparsest solution too hard
NP-hard

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Compressed sensing to the rescue

Convex relaxation

Recall $y = A\mathbf{x}^*$ where $\mathbf{x}^* \in \mathbb{R}^n$ is S -sparse

A : $k \times n$ random Gaussian matrix

Solve $\hat{\mathbf{x}} = \arg \min \|\mathbf{x}\|_1$ such that $A\mathbf{x} = \mathbf{y}$

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Can be solved fast

Solution will coincide with the sparsest $\mathbf{x}$ provided

A satisfies the restricted isometry property

$k > S \log n$

Another project idea

Ridge and Lasso

Already noted, brief review

Matrix Completion

This will be our segue into next topic: NLP
Also a chance to learn about singular values

Matrix completion

M is a matrix of user preferences
(Netflix challenge 480189×17770 , today much larger)
user i , movie j , rating in position M_{ij}
most observations unknown
observations in set Ω

Complete M using Ω

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Complete M using Ω
no reason this is this even possible!
Reminiscent of compressed sensing
inferring sparse signal with very few measurements
equivalent of sparsity?

“Sparsity” in matrices

Rank of a matrix, Singular Value Decomposition

Rank 1 matrices

General rank k matrices

Singular value decomposition

Outer product expression

Autoencoders

Singular value decomposition/Genres

$$M = U\Sigma V^T$$

Rank = number of non-zero singular values

Low rank r : few (r) singular values

$$M = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \cdots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

$\mathbf{u}_i$: how much does each person genre i ? $\mathbf{v}_i$: how would a person who likes genre i like each movie?

Matrix completion

Works when Ω is chosen uniformly at random

The rank of M is low

Singular value are incoherent with the standard basis
projecting the standard basis to the subspace
of singular vectors

Matrix completion: nuclear norm

$$X = \arg \min \|X\|_s$$

such that $X_{ij} = M_{ij}$, $ij \in \Omega$ where $\|X\|_s$ is the nuclear norm